

# Lending Game: Results in Round 1

## A. Simulation of the Market

Recall that for each loan application, your team is in competition with two other teams, which have different information sets than yours. If you are a type 2 lender (i.e., you have variable `signal2` for new applications) or a type 3 lender (i.e., you have variable `signal3`), the identity of the two teams that compete with you are different for each loan application because two teams are type 1 lenders. So we simulated the market by randomizing the identify of your type 1 competitor for each loan application. If you are a type 1 lender (i.e., you have variable `signal1` for new applications), the identity of the two teams that compete with you are the same for each loan application because there are only one type 2 lender and one type 3 lender participating to the game.

We also randomized loan applicants' types, that is, the lender for which each loan applicant is ready to pay 2% extra to borrow from this lender.

## B. Diagnostic Data

The detailed outcome of the market is in the file `Round1_Diagnostic_*.csv`. It contains the 100,000 loan applications. The variables in the dataset are (if you are a type 1 lender):

- `id`: Loan identifier.
- `own`: Interest rate offered by your team.
- `competing2`: Interest rate offered by lender 2. Missing interest rate means no offer.
- `competing3`: Interest rate offered by lender 3. Missing interest rate means no offer.
- `borrowertype`: =1,2,3 indicates the lender for which the borrower is ready to pay 2% extra.
- `accepted`: =1 if the borrower takes your loan offer; =0 if the borrower does not take your loan offer. Note that you can reconstruct this variable using the information about the three interest rates and the borrower's type.
- `default`: =1 if the borrower defaults on the loan; =0 if the borrower repays the loan.
- `profit`: Your profit on this loan. The profit if your loan offer is accepted is equal to interest rate times 10,000 if the borrower repays, and to negative 10,000 if the borrower defaults.

### C. League Table

The table below shows the market shares and the profits and losses (P&L) of all the teams participating to the game. The market share for each team is calculated as the market share on all loans for which it competes. The P&L for each team is the profit summed over all the loans made by the team.

|                 | Market share | P&L        |
|-----------------|--------------|------------|
| YaBalach        | 27.5%        | 5,765,940  |
| Too big to fail | 64.2%        | 2,423,710  |
| Et-Jo-Ro-Ul     | 30.1%        | -1,487,442 |
| Fiscal Fiasco   | 9.8%         | -3,129,429 |

### D. Humans vs. LLMs

Your team may have used help from an LLM. This is perfectly fine. Of course, your objective should be to add value relative to what LLMs can produce. To assess whether you added value relative to LLMs, we asked ChatGPT 5.4 to complete the assignment and re-simulated the market with GPT competing alongside the human teams. GPT ranked in the middle of the pack.

### E. Second Round

As you can see, it is challenging to turn profits in this game. So keep the team's spirits up and take round 2 as an opportunity to learn from the experience of round 1 and improve your strategy.

Your task for the second round is to use the diagnostic data to review your strategy and make offers to a new batch of loans. See the guidelines of the game for the detailed instructions. The simulation of profits in round 2 will be done in exactly the same way as in round 1. May the force be with you!